

SmartBasket AI Blueprint v5 – Master Expansion

This document is intended to replace and extend the previous blueprint as the master architectural specification for Claude. It focuses on removing ambiguity and defining production-grade systems while remaining achievable as a hackathon MVP.

1. Product Vision

SmartBasket becomes an AI grocery operating system rather than only a shopping comparison website. It combines retailer aggregation, intelligent recommendations, recipe intelligence, pantry management, nutrition planning, budgeting and meal generation into one platform.

2. Retail Data Platform

Maintain a canonical Master Product table with separate Store Listings for Coles, Woolworths, IGA, ALDI, Costco and future retailers. Store Listings contain live price, unit price, promotions, availability and historical pricing. Products never duplicate; store-specific information references the master product.

3. Scraping & ETL Pipeline

A scheduled scraper service periodically collects retailer data. Pipeline: Scraper → Parser → Cleaner → Barcode Matcher → Duplicate Detector → Nutrition Validator → Image Downloader → Price History Logger → Supabase. The website never scrapes directly; it only queries Supabase.

4. Cookbook Knowledge Base

Build a professional recipe database supporting breakfast, lunch, dinner, snacks, desserts, drinks, air fryer, meal prep, BBQ, vegan, vegetarian, gluten free, dairy free, keto, high protein, low calorie, student meals, budget meals, family meals and cuisine categories. Every recipe stores ingredients, quantities, steps, equipment, nutrition, allergens, substitutions, cost per serving, images, storage and leftovers.

5. Recommendation Engine

Ranking pipeline: hard filters (allergy, diet, availability) then weighted scoring (nutrition, protein per dollar, budget, recipe compatibility, user history, convenience). AI only explains deterministic results.

6. AI Architecture

Separate AI agents: Grocery Assistant, Recipe Planner, Budget Optimiser, Pantry Assistant, Nutrition Assistant, Product Comparator and Cooking Coach. Each has defined inputs, outputs, guardrails and deterministic validation.

7. Database

Expand schema to include Products, Brands, Categories, Stores, StoreListings, PriceHistory, Promotions, Recipes, RecipeIngredients, Pantry, PantryHistory, MealPlans, ShoppingLists, SavedJourneys, RecommendationEvents, SearchHistory, UserPreferences, Images and Admin tables.

8. Search

Implement semantic search with pgvector alongside keyword search. Users can search naturally (e.g. 'cheap high protein air fryer dinner').

9. Analytics

Track recommendation acceptance, basket savings, search behaviour, recipes cooked, pantry waste reduction and retailer comparisons.

10. Admin

Provide dashboards for recipe approval, scrape monitoring, duplicate resolution, price validation, analytics and user management.

11. Claude Build Rules

Claude must treat this blueprint as the source of truth, implement one phase at a time, keep all calculations deterministic, never invent nutritional facts, and ensure every feature is tested before proceeding.

Recommended Next Revision: Expand this into a 100+ page software requirements specification including ER diagrams, API contracts, UI wireframes, complete SQL schema, sequence diagrams, and implementation phases.